DAY 10 | EXCEL

My Data Analyst Journey

Logical Formulas:
IF, AND, OR, IFS & More

Day 10 of learning Excel in public.

IF AND OR TRUE

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Logical Test

The heart of every IF

What It Is

A condition that Excel checks, like "is the percentage 33% or more?"

The Answer

Always TRUE or FALSE (Boolean). Behind the scenes TRUE = 1 and FALSE = 0.

Example

Marks 50% against the 33% rule = TRUE, so the result is Pass.

Comparison Operators

Used to build every condition

=	Equal to
<>	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal

Example: =B2>=33% returns TRUE or FALSE. Text in a condition must be inside quotes, like =A1="Paid".

IF Function

Make Excel decide

Syntax

=IF(logical_test, value_if_true, value_if_false)

Pass / Fail

=IF(B2>=33%,"Pass","Fail")

Results

A 50% = Pass. B 30% = Fail. C 70% = Pass. D 20% = Fail. E 80% = Pass.

Text Rule

Text results go inside double quotes. Numbers and cell references do not.

IF Examples

Three real situations

`=IF(B2>=33%,"Pass","Fail")` — **Pass or Fail**

`=IF(C33="","Pending","Done")` — **Blank = Pending**

`=IF(A1>=100,A1+B1,B1-A1)` — **600**

Empty quotes "" mean a blank cell. If C33 is blank the answer is Pending, and if it holds 123 the answer is Done (like Payment Done). For the last one A1 = 100 and B1 = 500, so 100 + 500 = 600.

AND, OR & NOT

Combine more than one condition

AND

All conditions must be TRUE. Bonus: sales >= 50,000 AND rating >= 3.

OR

Any one condition is enough.

Formula

=AND(D24>=50000,E24>=3) gives TRUE or FALSE. OR is written the same way.

NOT

=NOT(TRUE) flips the answer to FALSE.

AND vs OR

Same data, different result

	AND	OR
A: 30,000, rating 4	FALSE	TRUE
B: 60,000, rating 5	TRUE	TRUE
C: 40,000, rating 2	FALSE	FALSE

Rule: sales >= 50,000 and rating >= 3. AND needs both to pass. OR needs only one.

IF with AND / OR

Nest AND or OR inside IF

IF + AND

```
=IF(AND(D24>=50000, E24>=3), "Bonus Eligible", "No Bonus")
```

IF + OR

```
=IF(OR(D24>=50000, E24>=3), "Bonus Eligible", "No Bonus")
```

How It Reads

AND or OR gives TRUE or FALSE, and IF turns that answer into words.

Nested IF

IF inside IF for many levels

Rating Rule

70,000+ = 5, 60,000+ = 4, 50,000+ = 3, 40,000+ = 2, otherwise 1.

Formula

=IF(D24>=70000, 5, IF(D24>=60000, 4, IF(D24>=50000, 3, IF(D24>=40000, 2, 1))))

Order Matters

Start from the highest condition. Excel stops at the first TRUE.

Brackets

Close one) for every IF you open. Here there are 4 IFs, so 4 closing brackets.

IFS Function

Many conditions, no nesting

IF vs IFS

IF checks one condition. IFS checks many conditions in a row.

Syntax

=IFS(test1, result1, test2, result2, ...)

Grade Example

=IFS(K4>=90%, "A", K4>=80%, "B", K4>=70%, "C", K4>=60%, "D", TRUE, "F")

The TRUE Trick

If every earlier test is FALSE, TRUE catches the rest and returns F. Without it you get #N/A.

Where to Use IFS

Real situations

Grades

Marks or percentage decide A+, A, B+ and so on. Longest scale first: 95% A+, 90% A, 85% A-...

Bonus %

Sales level decides what percentage of bonus someone earns.

Tax Slabs

Income level decides which tax percentage applies.

Remember

Put the highest condition first, and finish with TRUE for "everything else".

SWITCH Function

Match exact values

Syntax

=SWITCH(expression, value1, result1, value2, result2, ..., default)

Example

=SWITCH(A1,"A+",5000,"A",3000,"B",1000,0)

Result

Grade A+ gives 5000, A gives 3000, B gives 1000, anything else gives 0.

SWITCH vs IFS

SWITCH matches exact values only. IFS is for ranges like >= 80%.

IFERROR

Hide ugly errors

The Problem

50 / 0 gives #DIV/0!, which looks bad in reports.

The Fix

=IFERROR(F90/F91,"Value is Wrong") shows your message instead.

Why Useful

Keeps automation sheets and trackers clean.

IFNA

=IFNA(value, message) hides only #N/A. IFERROR hides every error, so use it carefully.

IS Functions

Check, then answer TRUE or FALSE

`=ISBLANK(C33)`	**Cell is empty**
`=ISNUMBER(A1)`	**Holds a number**
`=ISTEXT(A1)`	**Holds text**
`=ISERROR(A1)`	**Any error**
`=ISNA(A1)`	**Only #N/A**
`=ISLOGICAL(A1)`	**TRUE or FALSE**
`=ISFORMULA(A1)`	**Cell has a formula**
`=ISEVEN(50)`	**TRUE**

ISODD(7) also exists. These pair well with IF, like =IF(ISBLANK(C33),"Pending","Done").

Day 10 Shortcuts

Write big formulas easily

Accept the function name	Tab
Function Arguments box	Ctrl + A
Line break inside formula	Alt + Enter
Expand formula bar	Ctrl + Shift + U
Evaluate Formula	Alt + M + V
Show value of selected part	F9
Lock a reference	F4
Insert Function box	Shift + F3

Type =IF, press Tab to get =IF(, then press Ctrl+A. The dialog box lets you fill each part of a big formula.

Day 10 Cheat Sheet

All functions in one place

Decide

IF, IFS, SWITCH

Combine Conditions

AND, OR, NOT

Handle Errors

IFERROR, IFNA

Check Type

ISBLANK, ISNUMBER, ISTEXT, ISLOGICAL, ISERROR, ISNA, ISFORMULA, ISEVEN, ISODD

Operators

= <> > < >= <=

Day 10
Done!

IF, AND, OR, IFS, SWITCH
and IFERROR.

Can you grade marks with IFS?
Tell me in the comments.

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